```python
facts = {
    "fever",
    "cough"
}

rules = {
    "flu": ["fever", "cough"],
    "respiratory_infection": ["fever", "cough"],
    "cold": ["cough", "sneezing"]
}

def backward_chaining(goal):
    print("Checking:", goal)

    if goal in facts:
        print(goal, "is a known fact")
        return True

    if goal in rules:
        for condition in rules[goal]:
            if not backward_chaining(condition):
                print("Goal failed:", goal)
                return False

        print("Goal proved:", goal)
        return True

    print("No rule or fact found for:", goal)
    return False

print("BACKWARD CHAINING")
print("----------------------------")

goal = "flu"
print("Goal:", goal)

if backward_chaining(goal):
    print("\nConclusion: Flu is supported by the rules.")
else:
    print("\nConclusion: Goal cannot be proved.")
```

```
Python 3.13.1 (tags/v3.13.1:0671451, Dec  3 2024, 19:06:28) [MSC v.1942
64 bit (AMD64)] on win32
Type "help", "copyright", "credits" or "license()" for more information.
>>>
========== RESTART: C:/Users/user/Downloads/AIES.py ==========
========
BACKWARD CHAINING
----------------------------
Goal: flu
Checking: flu
Checking: fever
fever is a known fact
Checking: cough
cough is a known fact
Goal proved: flu

Conclusion: Flu is supported by the rules.
>>>
```